CREDITPATHAI – PROJECT DOCUMENTATION

1. Project Title

CreditPathAI: Intelligent Loan Risk Prediction and Decision Support System

2. Abstract

CreditPathAI is a full-stack machine learning-based application developed to predict the probability of loan default using borrower financial and demographic data. The system integrates advanced machine learning models with a modern web-based dashboard to provide real-time risk assessment and actionable insights.

The project is designed to assist financial institutions in evaluating loan applications efficiently by classifying borrowers into different risk categories such as Low, Medium, and High. Based on the predicted risk, the system provides recommendations such as approving the loan, reviewing it manually, or rejecting it.

The system consists of three major components:

- Machine Learning Model (XGBoost / LightGBM)
- FastAPI Backend for model serving
- React Frontend for visualization and interaction

Additionally, the system includes explainable AI features, allowing users to understand the reasoning behind predictions, along with interactive charts and historical tracking to enhance decision-making.

3. Objectives

The primary objectives of this project are:

- To develop a machine learning model capable of predicting loan default probability
- To classify borrowers into risk categories (Low, Medium, High)
- To design and implement a full-stack application integrating frontend and backend
- To provide actionable insights for financial decision-making
- To enhance interpretability using explainable AI techniques
- To create an interactive and user-friendly dashboard

4. System Architecture

The CreditPathAI system follows a layered architecture consisting of frontend, backend, and machine learning components.

- **Architecture Flow:**

User → React UI → Axios → FastAPI → ML Model → JSON Response → React UI

- **Detailed Explanation:**
  1. The user enters borrower details in the React frontend
  2. The frontend sends the data to the backend using Axios
  3. The FastAPI backend receives the request and processes it
  4. The trained machine learning model predicts the probability of default
  5. The backend classifies the risk and generates recommendations
  6. The response is returned in JSON format
  7. The frontend displays results along with charts and explanations

This architecture ensures scalability, modularity, and efficient communication between components.

5. Project Structure

```
creditpathai/
│
├── frontend/
│   ├── src/
│   │   ├── components/
│   │   │   ├── Form.js
│   │   │   ├── Dashboard.js
│   │   │   ├── Charts.js
│   │   │   ├── History.js
│   │   ├── App.js
│   ├── package.json
│
├── main.py
├── model.pkl
├── pipeline.pkl
├── utils.py
└── README.md
```

6. Technologies Used

Frontend Technologies:

- React.js (UI development)
- Axios (API communication)
- Plotly.js (data visualization)
- CSS (styling and layout)

Backend Technologies:

- FastAPI (API development)

- Uvicorn (server)
- Python

Machine Learning Libraries:

- Scikit-learn
- XGBoost
- LightGBM

Data Processing:

- Pandas
- NumPy

7. Dataset & Feature Engineering

The dataset contains borrower-related financial and demographic attributes used to predict loan default risk.

Input Features:

- Age
- Loan Amount
- Credit Score
- Months Employed
- Number of Credit Lines
- Interest Rate
- Loan Term
- Debt-to-Income Ratio

Feature Engineering:

To improve model performance, additional features were created:

- **Loan-to-Income Ratio**
- **Credit Utilization**
- **Interest Burden**
- **Employment Stability**

These features help the model better understand borrower behavior and financial risk.

8. Model Development

The model development process involved:

Data Preprocessing:

- Handling missing values
- Encoding categorical variables
- Feature scaling

Model Selection:

- XGBoost
- LightGBM

These models were chosen due to their ability to handle non-linear relationships and provide high accuracy.

Evaluation:

- ROC-AUC Score used for performance measurement

Output:

- Probability of loan default (0 to 1)

9. Backend Implementation

The backend is implemented using FastAPI to provide a RESTful API for prediction.

Endpoint:

POST /predict

Functionality:

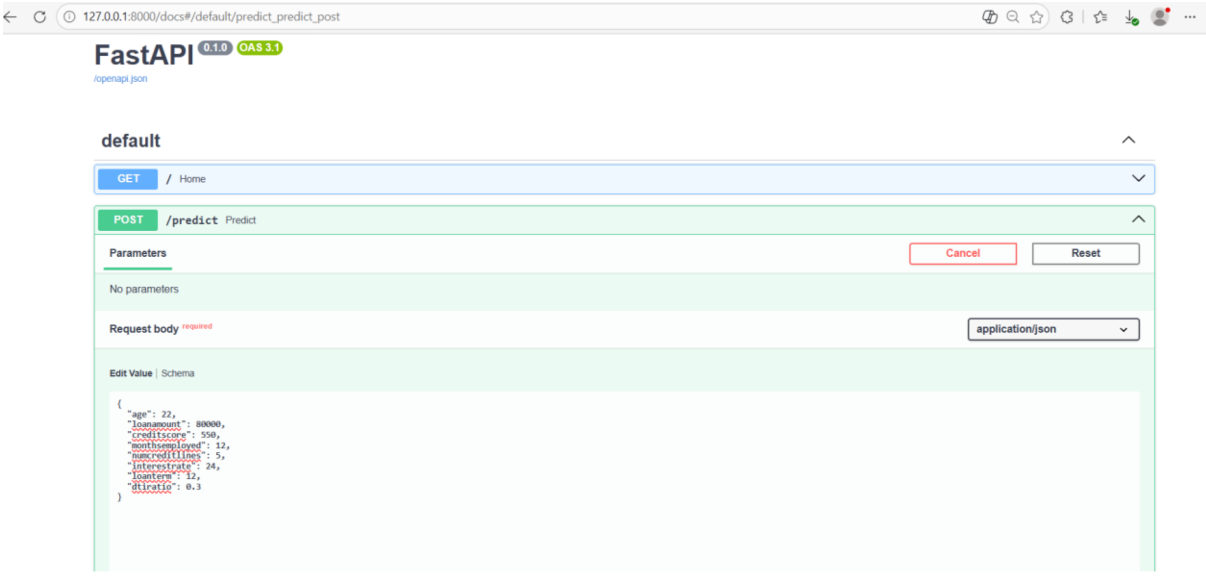
- Accepts borrower input data
- Performs feature engineering
- Uses trained model to predict probability
- Classifies risk level
- Returns structured JSON response

Output:

- Probability
- Risk Level
- Action Recommendation
- Reasons (Explainable AI)

Additional Features:

- CORS enabled for frontend integration
- Error handling using try-catch
- Fast response time



10. Frontend Implementation

The frontend is built using React and follows a component-based architecture.

Components:

- **Form Component**: Collects user input
- **Dashboard Component**: Displays results
- **Charts Component**: Visualizes data
- **History Component**: Stores past predictions

Features:

- Real-time API integration
- Dynamic UI updates
- Banking-style professional interface
- Responsive design

11. Data Visualization

To enhance interpretability, the system uses Plotly charts:

- **Gauge Chart** → Displays risk probability
- **Bar Chart** → Shows risk distribution

These charts provide intuitive understanding of predictions.

12. Risk Classification Logic

Probability Range	Risk Level	Action
< 0.25	Low	Approve Loan
0.25 – 0.50	Medium	Review Manually
> 0.50	High	Reject Loan

13. Explainable AI

The system includes explainability by providing reasons for predictions:

- Low Credit Score
- High Debt-to-Income Ratio
- Low Employment Stability
- High Loan Amount

This increases transparency and helps users trust the model's decisions.

14. Additional Features

- Prediction history using localStorage
- Filtering based on risk levels
- Icon-based explanation system
- Interactive charts

15. Results

The system successfully:

- Predicts loan default probability
- Classifies borrowers into risk categories
- Provides actionable recommendations
- Displays results with charts and explanations

The backend processes requests efficiently, and the frontend presents data clearly.

16. Testing (User Acceptance Testing)

Testing was conducted using multiple input scenarios:

- Low-risk inputs
- Medium-risk inputs
- High-risk inputs

Verified:

- Correct API response
- Accurate risk classification
- Proper UI rendering
- Chart updates

17. Challenges Faced

- Handling CORS issues
- Integrating frontend with backend
- Debugging API responses
- Tuning risk classification thresholds
- Designing a professional UI

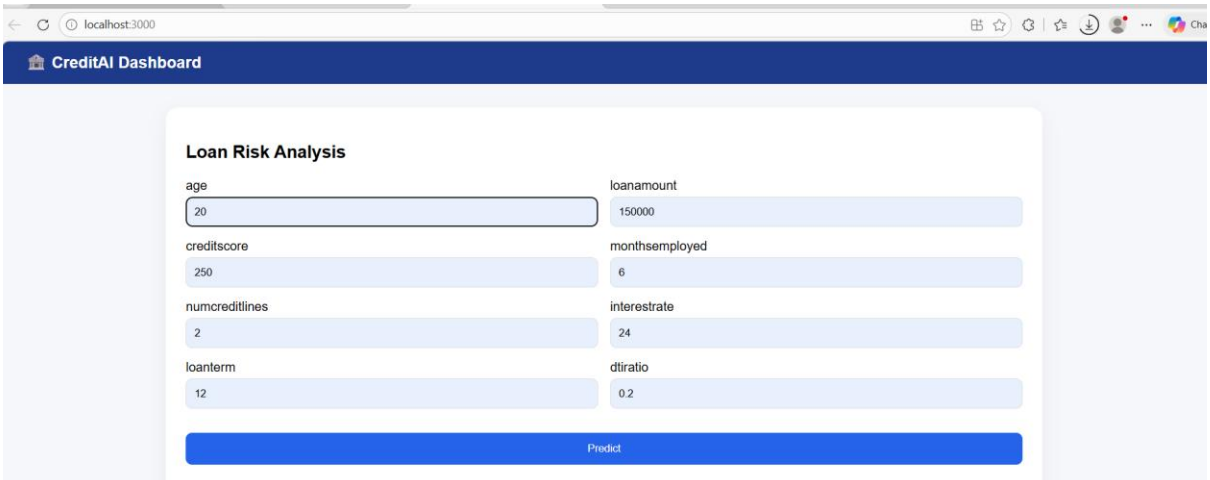
18. Future Scope

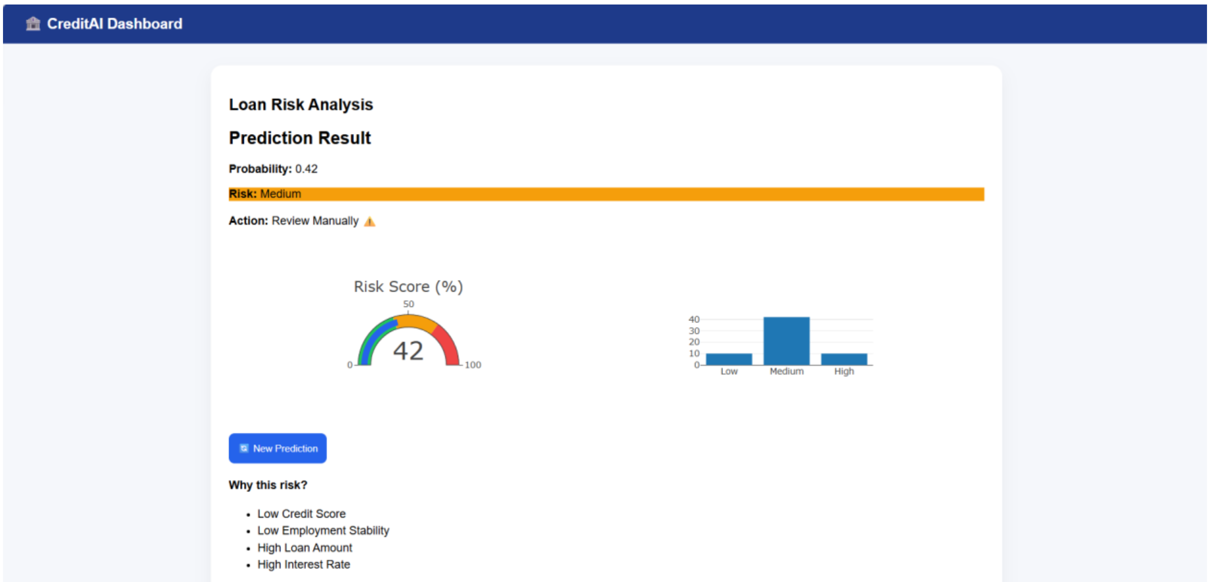
- Deploy system on cloud platforms
- Integrate with real banking systems
- Add authentication and security
- Improve model accuracy
- Use deep learning techniques

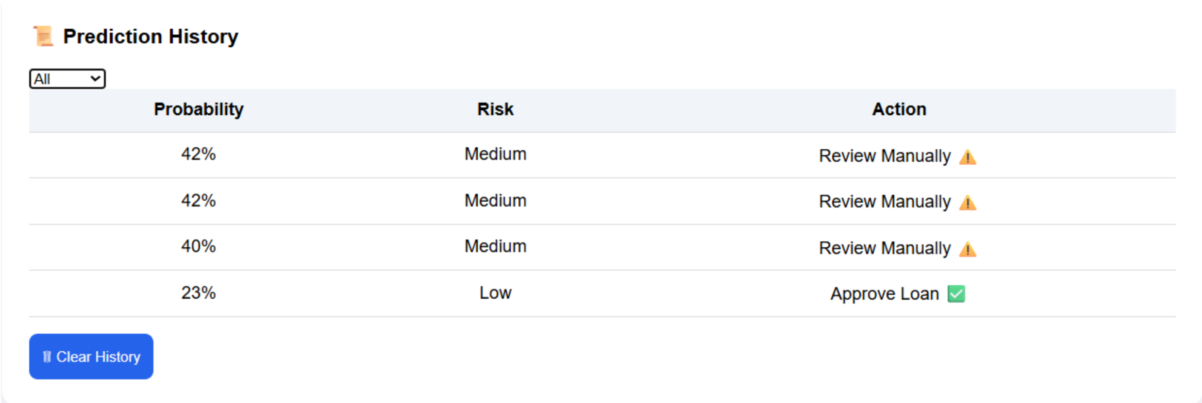
19. Conclusion

CreditPathAI demonstrates a complete full-stack machine learning solution for loan risk prediction. The system not only provides accurate predictions but also enhances decision-making through visualization and explainability. It can be extended into real-world financial applications.

RESULT:


localhost:3000
CreditAI Dashboard
Loan Risk Analysis
age
20
loanamount
150000
creditscore
250
monthsemployed
6
numcreditlines
2
interestrate
24
loanterm
12
dtiratio
0.2
Predict


CreditAI Dashboard
Loan Risk Analysis
Prediction Result
Probability: 0.42
Risk: Medium
Action: Review Manually
Risk Score (%)
50
42
0
100
40
30
20
10
0
Low
Medium
High
New Prediction
Why this risk?
Low Credit Score
Low Employment Stability
High Loan Amount
High Interest Rate


Prediction History
All
Probability
Risk
Action
42%
Medium
Review Manually
42%
Medium
Review Manually
40%
Medium
Review Manually
23%
Low
Approve Loan
Clear History